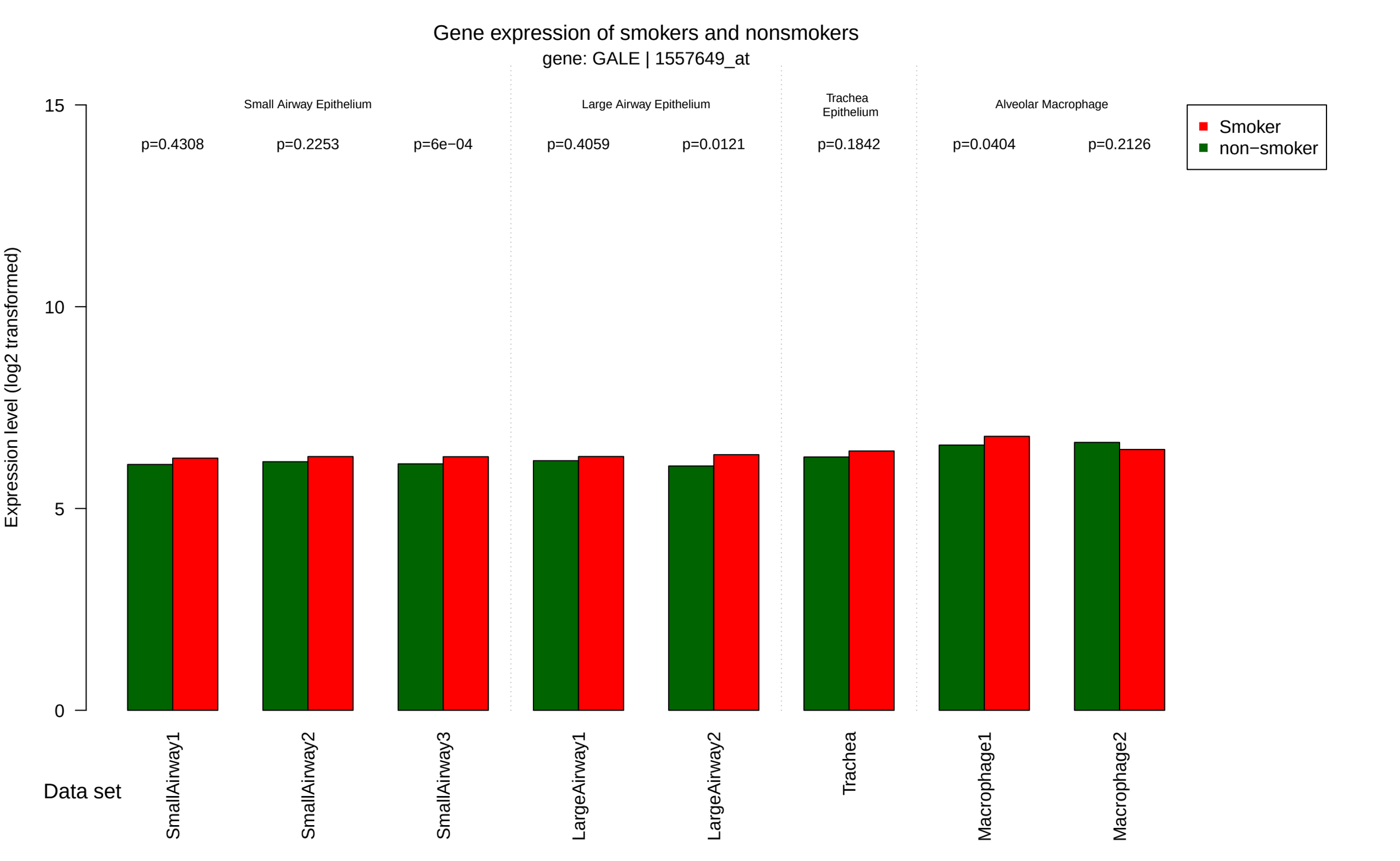
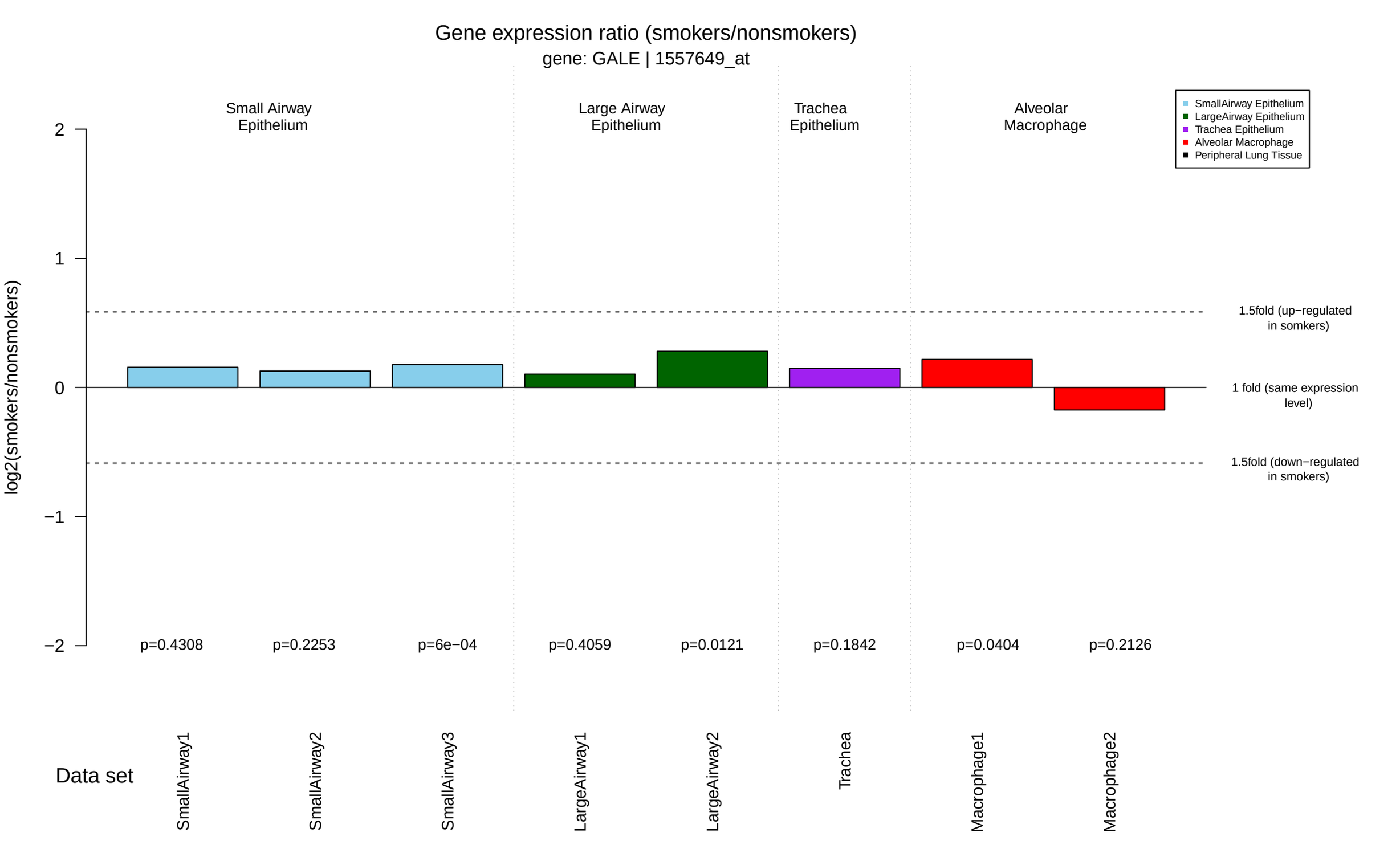
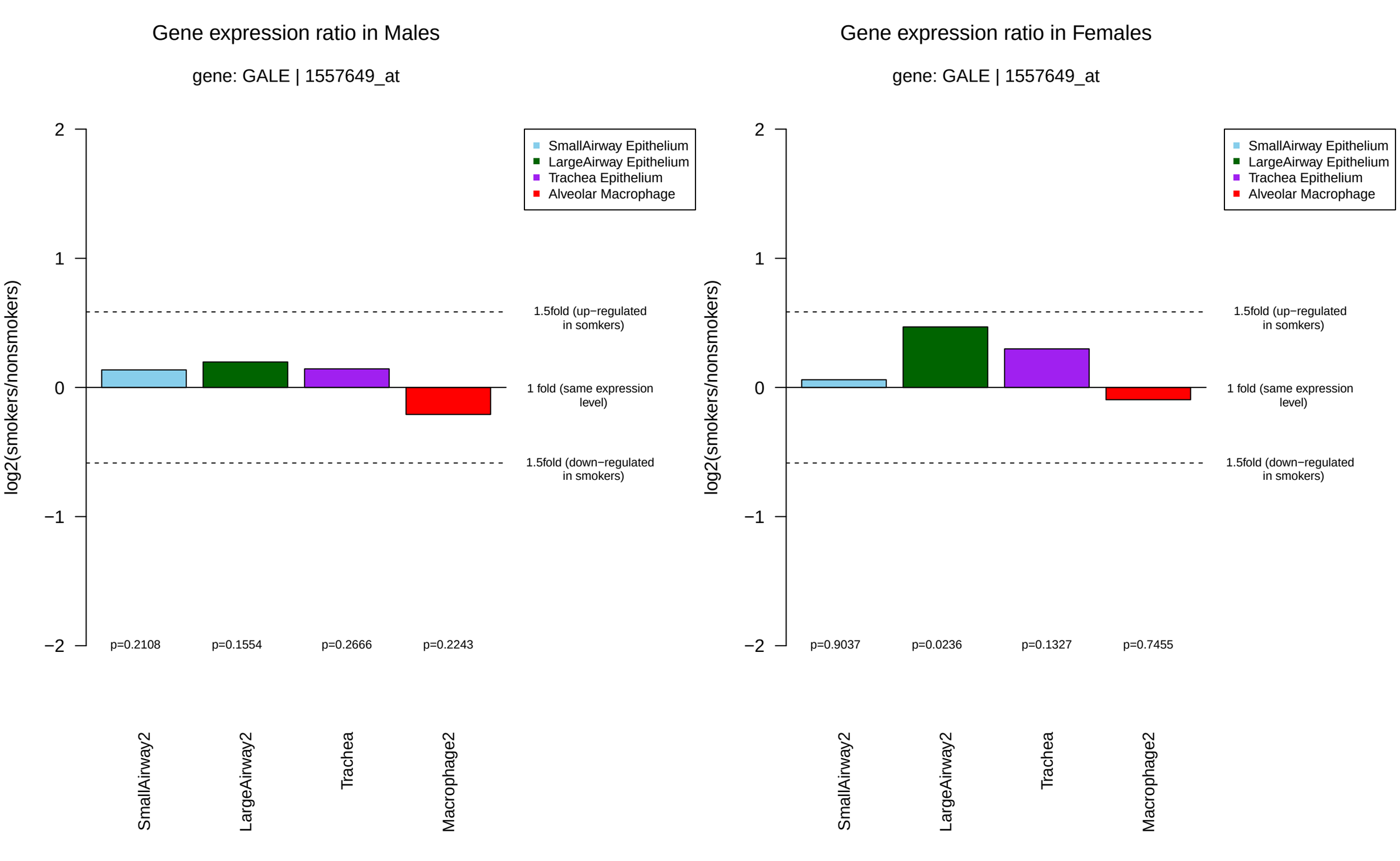
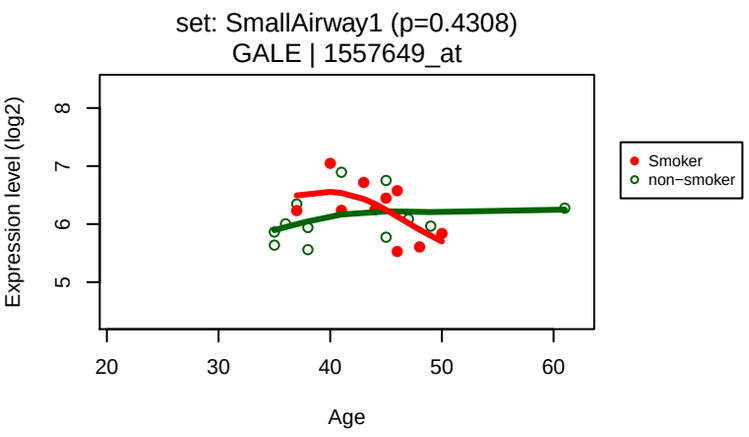
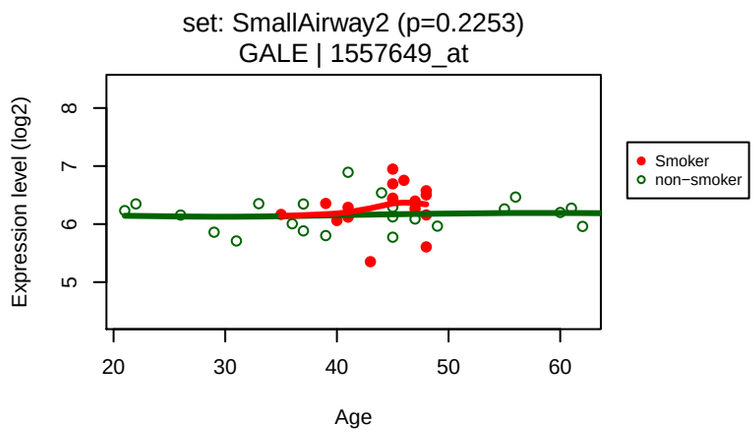
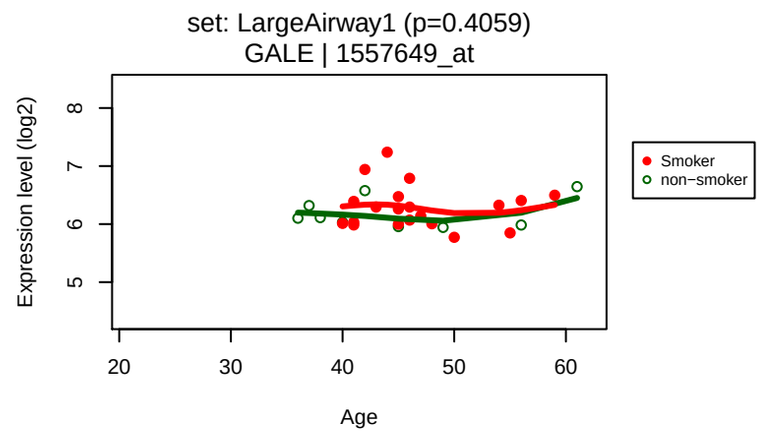

Gene expression of smokers and nonsmokers
gene: GALE | 1557649_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4308
p=0.2253
p=6e−04
p=0.4059
p=0.0121
p=0.1842
p=0.0404
p=0.2126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

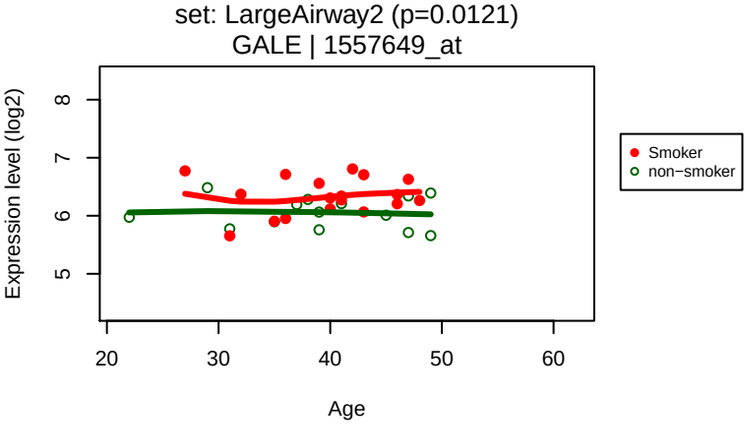

Gene expression ratio (smokers/nonsmokers)
gene: GALE | 1557649_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4308
p=0.2253
p=6e−04
p=0.4059
p=0.0121
p=0.1842
p=0.0404
p=0.2126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

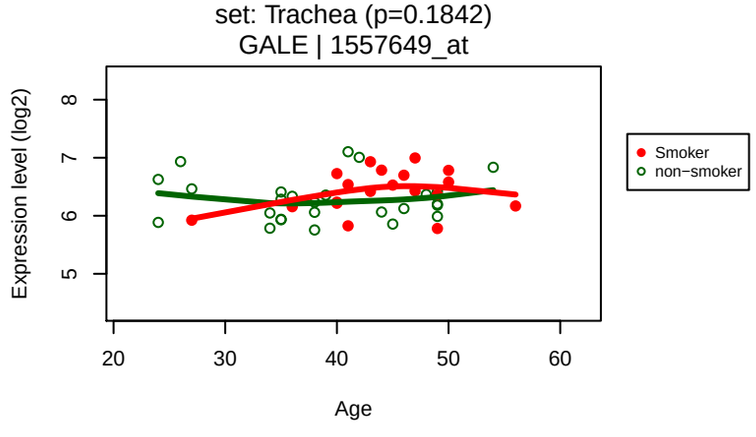

Gene expression ratio in Males
gene: GALE | 1557649_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2108
p=0.1554
p=0.2666
p=0.2243
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GALE | 1557649_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9037
p=0.0236
p=0.1327
p=0.7455
SmallAirway2
LargeAirway2
Trachea
Macrophage2

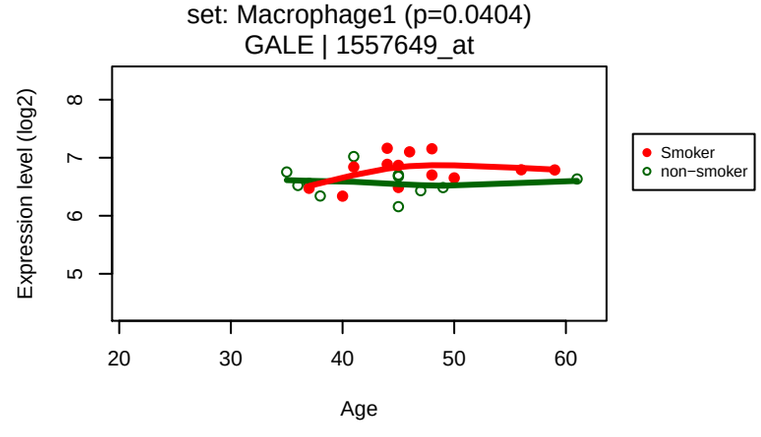

set: SmallAirway1 (p=0.4308)
GALE | 1557649_at
Expression level (log2)
Age
Smoker
non−smoker

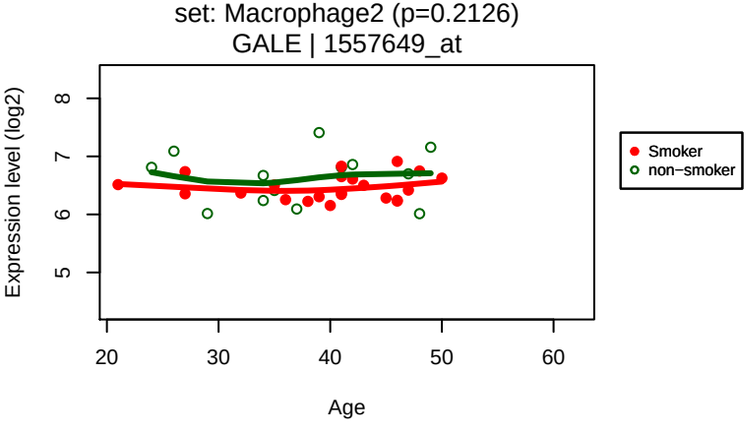

set: SmallAirway2 (p=0.2253)
GALE | 1557649_at
Expression level (log2)
Age
Smoker
non−smoker

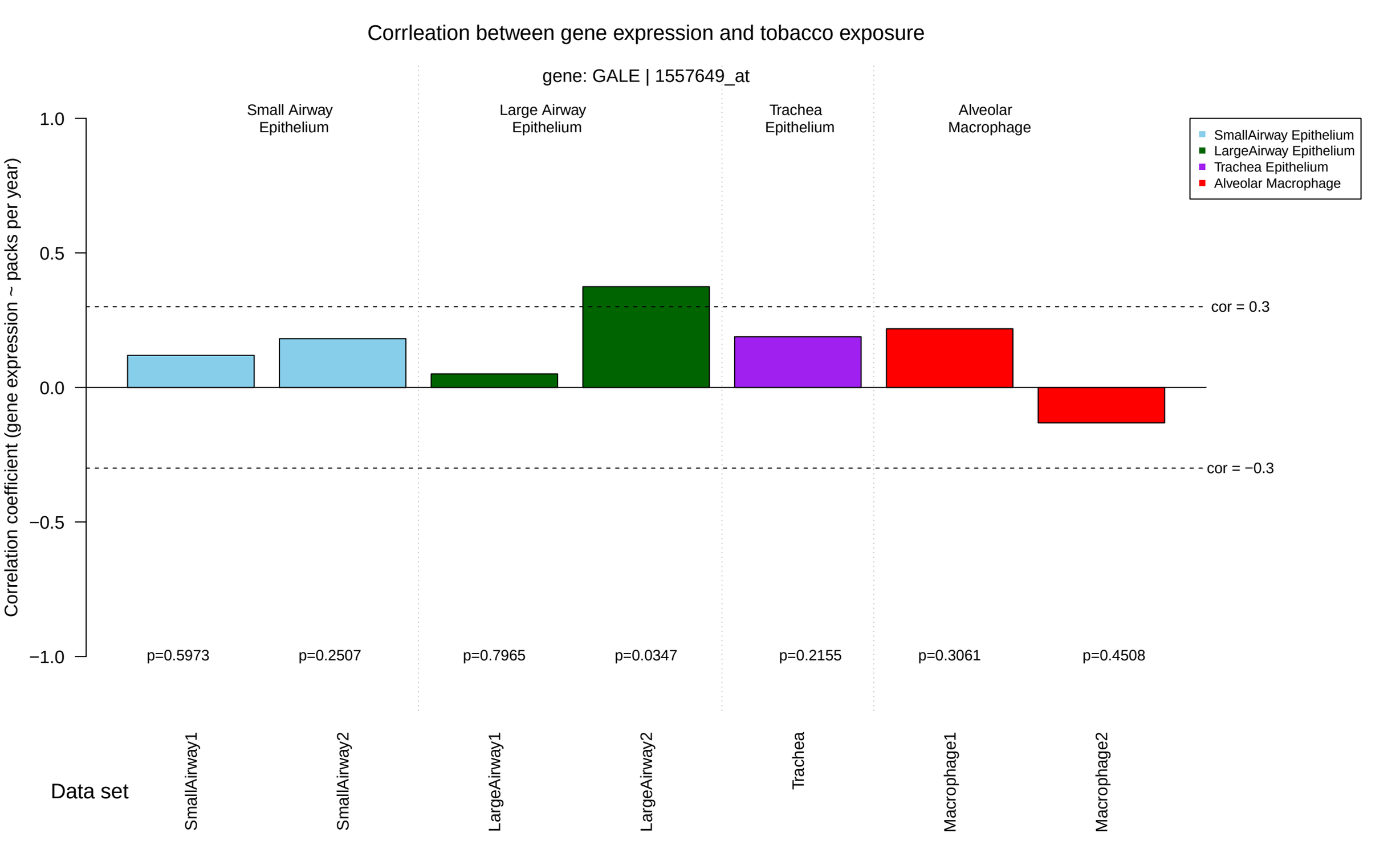

set: LargeAirway1 (p=0.4059)
GALE | 1557649_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0121)
GALE | 1557649_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1842)
GALE | 1557649_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0404)
GALE | 1557649_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2126)
GALE | 1557649_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GALE | 1557649_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5973
p=0.2507
p=0.7965
p=0.0347
p=0.2155
p=0.3061
p=0.4508
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2